

Focus Friend
MINOR PROJECT REPORT

**Submitted in partial fulfillment of the requirement for the Degree of
Bachelors of Engineering in Computer Science & Engineering**

Submitted To: Parul University



PARUL UNIVERSITY, VADODARA, GUJARAT (INDIA)

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

PARUL INSTITUTE OF TECHNOLOGY VADODARA, GUJARAT

SESSION: AY 2025-2026

Parul University Parul Institute of Technology



(Session: 2025 -2026)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE

This is to certify that (“**Jiwan Chaudhary, V Naik Surya, Shivam Ingle, Shubham Girawale**”) Students of **CSE VI Semester** of “ **Parul Institute of Technology, Vadodara**” has completed their **Minor Project** titled “**Focus Friend**”, as per the syllabus and has submitted a satisfactory report on this project as a partial fulfillment towards the award of degree of **Bachelor of Technology in Computer Science and Engineering** under Parul University, Vadodara, Gujarat (India).

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MINOR PROJECT EVALUATION

Evaluated by Project Supervisor–Evaluation done out of 100 Marks – Weightage will be as per scheme Marks.

Name of Student: _____ Enrolment No. _____

Sl. No.	Criteria	Max Marks	Marks Awarded by Supervisor	Remarks
1	Statement of the Research problem (Topic)	10		
2	Aim, Objectives and Methodology	20		
3	Justification of the problem	10		
4	Summary of Critical Evaluation of Journal Papers (5 Journal papers minimum)	25		
5	Presentation and communication skills	15		
6	Relevance and quality of PPT slides	10		
7	References	10		
Total		100		

Remarks (Justification for the marks awarded)

Name and signature of the Project Supervisor with date

ANNEXURE - IV



PARUL UNIVERSITY
FACULTY OF ENGINEERING AND TECHNOLOGY
ENGINEERING DEPARTMENT

Student Evaluation Card (6th semester)

Project Group Identification	
Project Title	
Type of Project	
Name of Project Supervisor	

SR NO	NAME OF THE STUDENTS	ENROLLMENT NO. STUDENTS	DEPARTMENT
1			
2			
3			
4			



PARUL UNIVERSITY
FACULTY OF ENGINEERING AND TECHNOLOGY
ENGINEERING DEPARTMENT

Date: _____

MID SEMESTER PROJECT EVALUATION – 6th Semester

Sr. No.	Comments given by Internal Review Panel	Modification done based on Comments (To be filled by Supervisor)
Particulars	Internal Review Panel	
	Expert 1	Expert 2
Name:		
sign		

PARUL INSTITUTE OF TECHNOLOGY
PARUL UNIVERSITY

Minor Project Report
FOCUS FRIEND

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Department: B.Tech Cyber Security

Submission Date: March 2026

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1. Abstract

Focus Friend is an intelligent web-based academic productivity system designed to help students organize their study materials, track academic progress, and validate their understanding through AI-assisted learning tools. Many students face challenges in managing subjects, keeping track of chapter completion, and verifying whether they have actually understood the material they study. Traditional systems usually focus only on storing documents rather than ensuring learning outcomes. This project addresses these issues by combining study management, document processing, artificial intelligence assistance, and knowledge validation into a single integrated platform.

The system allows users to create subjects and chapters, upload chapter-wise PDF study materials, and read them directly inside the application. A key feature of the system is its AI-powered assistant which enables users to ask questions directly from the uploaded documents. The system uses document parsing and semantic search techniques to extract relevant content and generate context-aware answers. This ensures that the responses provided by the AI are grounded in the student's own study material rather than generic information.

Another important feature of the system is its quiz-based chapter completion mechanism. The system automatically generates quizzes from the uploaded PDF content to test the student's understanding. A chapter is marked as complete only when the student successfully answers all quiz questions correctly. This approach promotes active learning and ensures mastery-based progress tracking instead of passive reading.

The backend of the application is developed using Node.js and Express.js, while MongoDB is used as the database for storing user data, subjects, chapters, and extracted PDF content. The frontend is built using HTML, CSS, and JavaScript to provide a simple and responsive user interface. AI capabilities are integrated using the Google Gemini API. Security features such as JWT authentication, password hashing, input validation, and rate limiting are implemented to ensure secure access and data protection.

The expected outcome of this project is to provide students with a structured learning environment that improves study discipline, enhances understanding, and provides measurable academic progress. The project demonstrates the practical integration of modern web technologies, artificial intelligence, document processing, and security practices to build a real-world educational productivity solution.

2.Introduction

In the modern educational environment, students deal with a large amount of digital study material such as lecture notes, reference books, PDFs, and online resources. Although digital resources are easily available, students often face difficulties in organizing these materials, tracking their study progress, and verifying whether they truly understand the topics they study. Most existing study tools only provide storage facilities but do not provide mechanisms to validate learning or measure knowledge acquisition.

Traditional learning methods often involve passive reading, where students read study material without testing their understanding. This leads to poor retention of knowledge and ineffective study habits. Furthermore, when students have doubts, they often use general AI tools or search engines that may not always provide answers based on their specific study material. This creates a gap between study organization and actual learning validation.

Focus Friend is designed to address these challenges by providing an integrated academic productivity platform that combines study organization, document management, artificial intelligence assistance, and progress tracking. The system allows students to create subjects and divide them into chapters, upload study material in PDF format, and interact with the content through AI-based question answering. This makes the learning process more interactive and structured.

The system also introduces a quiz-based validation mechanism to ensure that students do not just read the material but also understand it. Before a chapter is marked as completed, the system generates a quiz based on the uploaded study material. Students must achieve full marks to mark the chapter as complete. This ensures mastery-based learning and encourages active engagement with study material.

From a technical perspective, the application follows a client-server architecture. The frontend is developed using HTML, CSS, and JavaScript to provide a responsive and user-friendly interface. The backend is implemented using Node.js and Express.js to handle business logic and API communication. MongoDB is used as the database to store user information, subjects, chapters, and document data. The system also uses PDF parsing libraries to extract text from uploaded documents and integrates Google Gemini AI to provide intelligent question answering and quiz generation.

Security is an important aspect of the system. The application implements JWT-based authentication for secure session management, password hashing for protecting user credentials, request validation to prevent malicious input, and rate limiting to prevent abuse of system resources. These measures ensure that the platform follows basic cybersecurity best practices while handling user data.

The Focus Friend system aims to transform the traditional study process into a structured and measurable learning workflow. By combining document interaction, AI assistance, and progress validation, the system helps students become more disciplined learners. The

project demonstrates how modern web technologies and artificial intelligence can be integrated to solve real-world educational problems and improve the effectiveness of digital learning systems.

3. Aim and Objectives

3.1 Aim

The main aim of the *Focus Friend* project is to develop an intelligent study management and academic assistance system that helps students organize their subjects, manage chapter-wise study materials, interact with uploaded PDFs using artificial intelligence, and verify their understanding through automated quiz-based evaluation. The system is designed to improve study discipline, promote active learning, and provide measurable academic progress through a structured digital learning environment.

The project also aims to demonstrate the practical implementation of modern web technologies combined with artificial intelligence to solve real-world educational challenges. By integrating document processing, AI-based question answering, and progress tracking into a single platform, the system provides a smart learning assistant that goes beyond traditional document storage systems.

Additionally, the project focuses on implementing secure authentication and data protection mechanisms such as JWT authentication, password hashing, and input validation to ensure that user data remains protected. The aim is not only to build a functional academic tool but also to demonstrate good software engineering and cybersecurity practices.

3.2 Objectives

The main objectives of the Focus Friend system are:

1. Study Material Organization

To provide a structured platform where students can create subjects and manage chapter-wise study materials in an organized manner.

2. PDF Management System

To allow students to upload chapter-specific PDF documents and read them directly inside the application without needing external software.

3. AI-Based Question Answering

To implement an AI assistant that can answer questions based on the uploaded study material, ensuring context-aware and relevant responses.

4. Knowledge Validation through Quizzes

To develop a quiz generation system that creates questions from uploaded PDFs and evaluates student understanding before allowing chapter completion.

5. Progress Tracking System

To create a dashboard that shows total subjects, total chapters, completed chapters, and overall completion percentage to help students monitor their learning progress.

6. Secure Authentication System

To implement secure login and registration functionality using JWT authentication and password encryption techniques.

7. Document Processing Implementation

To extract and process text from uploaded PDFs using parsing libraries for AI interaction and semantic search functionality.

8. User-Friendly Interface

To design a simple and responsive user interface that allows easy navigation and interaction with study resources.

9. Practical Implementation of AI Integration

To demonstrate how artificial intelligence can be integrated into web applications to improve learning efficiency and academic productivity.

10. Scalable System Design

To design the system in a modular way so that future improvements such as flashcards, summaries, and advanced analytics can be easily added.

4. Problem Statement

In today's digital learning environment, students have access to a large amount of study material in the form of PDFs, lecture notes, presentations, and online resources. However, despite the availability of these resources, students often struggle to manage their study content effectively. Most students store their materials in different folders or platforms without a proper tracking mechanism, which leads to poor organization and inefficient study practices.

Another major problem is the lack of progress tracking. Students usually do not have a proper system to monitor which chapters they have completed and which topics still require attention. Traditional study methods rely heavily on self-discipline, and there is no automated system to verify whether a student has actually completed or understood a chapter. As a result, students may assume they have completed a topic without properly testing their understanding.

Students also face challenges when they have doubts related to their study material. Although general AI tools and search engines are available, they usually provide answers based on general internet knowledge rather than the student's own study documents. This creates a gap between the student's specific learning content and the help they receive. A system that can provide answers directly from the student's own uploaded material would significantly improve the learning process.

Another limitation of existing systems is the absence of knowledge validation. Many study management applications allow students to mark chapters as complete manually without any testing mechanism. This does not guarantee that the student has actually understood the material. A proper learning system should include a validation process such as quizzes or assessments before marking progress as complete.

The following table summarizes the key problems identified:

Problem Area	Description
Poor Organization	Study materials are scattered across multiple locations without structure
No Progress Tracking	No clear method to measure chapter completion
Passive Learning	Students read without testing their understanding
Lack of Contextual AI Help	AI tools do not use the student's own documents
No Knowledge	Chapters can be marked complete without assessment

Problem Area

Description

Validation

Resource Fragmentation Study tools and learning tools are separated

Considering these challenges, there is a strong need for an integrated system that can:

- Organize study materials in a structured manner
- Track academic progress automatically
- Allow interaction with study documents
- Provide AI-based assistance grounded in study material
- Validate learning through quizzes before marking completion

The Focus Friend system is proposed as a solution to these problems. The system integrates study organization, document interaction, artificial intelligence assistance, and knowledge validation into a single platform. By combining these features, the system aims to improve learning efficiency, promote disciplined study habits, and provide measurable academic progress.

The problem addressed by this project is therefore the absence of a unified academic productivity system that not only manages study materials but also verifies understanding and promotes effective learning practices through intelligent assistance and structured progress tracking.

5. Proposed Methodology

The Focus Friend system follows a structured client-server architecture where the frontend interacts with the backend through REST APIs, and the backend communicates with the database and AI services. The methodology focuses on secure user management, document processing, AI interaction, and knowledge validation through quizzes. The system workflow is designed to ensure that students not only store study material but also actively engage with it.

5.1 User Authentication Module

The first step in the system is secure user authentication. The user must register before accessing the system features. During registration, the system collects user details such as email and password. Passwords are securely stored using hashing techniques to prevent unauthorized access.

After registration, the user logs into the system using valid credentials. JWT (JSON Web Token) authentication is used to manage user sessions securely. The token is generated after successful login and is used to authorize further requests.

This module includes:

- User registration
- Secure login system
- Password hashing
- JWT token generation
- Session validation
- Profile verification

This ensures that only authorized users can access the system.

5.2 Subject and Chapter Management

After successful authentication, users can create subjects based on their academic needs. Each subject can contain multiple chapters. When a subject is created, the system automatically generates chapter entries based on the number specified by the user.

This structured approach helps students divide their study material into manageable units.

This module includes:

- Subject creation
- Subject renaming

- Subject deletion
- Automatic chapter generation
- Chapter editing
- Chapter progress tracking

This helps in maintaining a well-organized academic structure.

5.3 PDF Upload and Processing

The system allows users to upload PDF documents related to specific chapters. After uploading, the system stores the document securely on the server and processes it using a PDF parsing library.

The system extracts:

- Full document text
- Page-wise text
- Structured content chunks

This extracted content is stored in the database for AI processing and semantic search.

This module includes:

- PDF upload functionality
- File storage management
- Text extraction from PDF
- Page segmentation
- Document indexing

This step prepares the document for intelligent interaction.

5.4 AI-Based Question Answering System

After PDF processing, the user can interact with the uploaded document through an AI assistant. When a user asks a question, the system searches relevant text from the extracted document content using semantic matching techniques.

The relevant content is then sent to the AI model to generate a contextual answer. This ensures that responses are based on the student's own study material rather than general knowledge.

This module includes:

- AI question interface
- Semantic text search
- Context extraction
- AI response generation
- Document-grounded answers

This feature improves learning by providing personalized assistance.

5.5 Quiz Generation and Knowledge Validation

To ensure that students understand the material, the system generates quizzes from the uploaded PDF content. The quiz typically consists of multiple questions based on chapter content.

Students must answer all questions correctly to mark a chapter as complete. If the student fails, they must retry the quiz. This ensures mastery-based learning rather than superficial completion.

This module includes:

- Automatic quiz generation
- Answer evaluation
- Score calculation
- Chapter completion validation
- Retry mechanism

This step ensures learning verification.

5.6 Dashboard and Progress Monitoring

The system provides a dashboard where students can monitor their academic progress. The dashboard shows important statistics such as total subjects, total chapters, completed chapters, and completion percentage.

This visual feedback helps students understand their study progress and motivates them to complete remaining chapters.

Dashboard features include:

- Subject count display
- Chapter completion tracking
- Progress percentage calculation

- Learning status visualization

This improves academic discipline and productivity.

5.7 System Security Implementation

Security is an important part of the methodology. The system implements multiple security measures to protect user data and prevent misuse.

Security mechanisms include:

- JWT authentication
- Password hashing
- Input validation
- Rate limiting
- Secure API access
- Protected routes

These measures ensure the confidentiality and integrity of user data.

5.8 System Workflow Summary

The overall workflow of the system follows this sequence:

User Registration/Login
→ Subject Creation
→ Chapter Generation
→ PDF Upload
→ Text Extraction
→ AI Question Answering
→ Quiz Generation
→ Quiz Evaluation
→ Chapter Completion
→ Dashboard Update

This structured workflow ensures proper learning management and validation.

6. Project Modules

The Focus Friend system is divided into several functional modules to ensure modular development, easy maintenance, and scalability. Each module is responsible for a specific functionality such as authentication, subject management, document processing, AI interaction, quiz validation, and progress monitoring. The modular architecture helps in separating responsibilities and improving system efficiency.

6.1 Authentication Module

The Authentication Module is responsible for managing user identity and secure access to the system. This is one of the most important modules because it ensures that only authorized users can access the application features.

The module provides user registration and login functionality. During registration, user data such as email and password are collected. Passwords are stored using secure hashing techniques to protect sensitive information. After successful login, the system generates a JWT token which is used for session management and API authorization.

Main functionalities:

- User registration
- User login
- Email verification
- Password hashing
- JWT token generation
- Session management
- Protected routes
- User profile verification

Security features implemented:

- Encrypted password storage
- Token-based authentication
- Request validation
- Unauthorized access prevention

This module ensures data security and controlled access to the system.

6.2 Subject Management Module

The Subject Management Module allows users to organize their academic structure by creating subjects. This module helps students manage different courses or topics efficiently.

Users can create new subjects, rename existing subjects, and delete subjects if required. When a subject is created, the system automatically generates chapters based on the number entered by the user. This reduces manual effort and ensures structured organization.

Main functionalities:

- Add new subject
- Edit subject name
- Delete subject
- Auto generation of chapters
- Subject listing
- Subject progress tracking

Advantages:

- Better academic organization
- Easy navigation of study content
- Logical subject structure
- Improved study planning

This module forms the foundation of the study organization system.

6.3 Chapter Management Module

The Chapter Management Module helps users manage individual chapters within subjects. Each chapter represents a unit of study and can be linked with a specific PDF document.

Users can rename chapters, monitor completion status, and track progress. The chapter is marked as completed only after successful quiz completion, ensuring knowledge validation.

Main functionalities:

- Chapter creation (auto-generated)
- Chapter renaming
- Chapter completion tracking

- Linking chapters with PDFs
- Chapter status monitoring

Benefits:

- Structured learning units
- Clear completion tracking
- Knowledge validation before completion
- Better progress visibility

This module ensures systematic chapter-level study tracking.

6.4 PDF Management Module

The PDF Management Module handles document upload, storage, and processing. This module is responsible for preparing study material for AI interaction.

When a user uploads a PDF, the system stores the file and extracts text using a PDF parsing library. The extracted text is divided into chunks and stored for semantic search operations. Users can also read the PDF directly inside the application.

Main functionalities:

- PDF upload
- File storage
- Text extraction
- Page-wise content separation
- PDF viewer integration
- Page navigation
- Document linking with chapters

Technical processes involved:

- File handling using Multer
- Text extraction using PDF parsing libraries
- Content chunking for AI search
- Database storage of extracted text

This module enables document interaction and AI processing.

6.5 AI Assistant Module

The AI Assistant Module provides intelligent interaction with study material. This module allows students to ask questions related to their uploaded PDFs.

When a question is asked, the system searches relevant text from stored document chunks using semantic similarity techniques. The relevant content is then provided to the AI model to generate a contextual response.

Main functionalities:

- Question input interface
- Context retrieval
- AI response generation
- Document-aware answers
- Page-aware responses
- Context filtering

Advantages:

- Personalized learning assistance
- Faster doubt solving
- Context-based responses
- Improved understanding of topics

This module makes the system intelligent and interactive.

6.6 Quiz Module

The Quiz Module is responsible for knowledge validation. This module ensures that students actually understand the study material before marking a chapter as complete.

The system generates quiz questions automatically from the uploaded PDF content. Students must answer all questions correctly to complete the chapter. If they fail, they must retake the quiz.

Main functionalities:

- Automatic quiz generation
- Multiple question generation
- Answer validation
- Score calculation
- Quiz retry option
- Chapter completion unlocking

Advantages:

- Encourages active learning
- Validates understanding
- Prevents false completion
- Promotes mastery learning

This module is critical for ensuring educational effectiveness.

6.7 Dashboard Module

The Dashboard Module provides a visual representation of student progress. It shows important academic metrics to help students monitor their learning journey.

The dashboard provides statistical summaries such as total subjects, total chapters, completed chapters, and completion percentage.

Main functionalities:

- Total subjects display
- Total chapters display
- Completed chapters count
- Progress percentage calculation
- Learning progress visualization
- Status indicators

Benefits:

- Motivates students
- Improves study discipline
- Provides measurable progress
- Helps identify pending work

This module provides performance insights.

6.8 Database Module

The Database Module manages storage of all system data. MongoDB is used as the database because of its flexibility and ability to store structured and semi-structured data.

Stored data includes:

- User data
- Subject data
- Chapter data
- PDF metadata
- Extracted text chunks
- Quiz data
- Verification tokens

Advantages of MongoDB:

- Flexible schema design
- JSON-based storage
- Easy scalability
- Fast retrieval

This module ensures reliable data management.

6.9 Security Module

The Security Module ensures that the system follows basic cybersecurity practices to protect user data and prevent misuse.

Security implementations include:

- JWT authentication
- Password hashing (bcrypt)
- Input validation
- Rate limiting
- Secure API routes
- Error handling
- Token verification

Security benefits:

- Protection from unauthorized access
- Data confidentiality
- Prevention of brute force attacks
- Safe API communication

This module ensures trust and reliability of the system.

Module Interaction Summary

All modules work together to create a complete academic productivity system:

Authentication Module → Controls access

Subject Module → Organizes subjects

Chapter Module → Tracks study units

PDF Module → Processes documents

AI Module → Provides assistance

Quiz Module → Validates learning

Dashboard Module → Shows progress

Security Module → Protects system

This modular structure ensures scalability, maintainability, and efficient functionality.

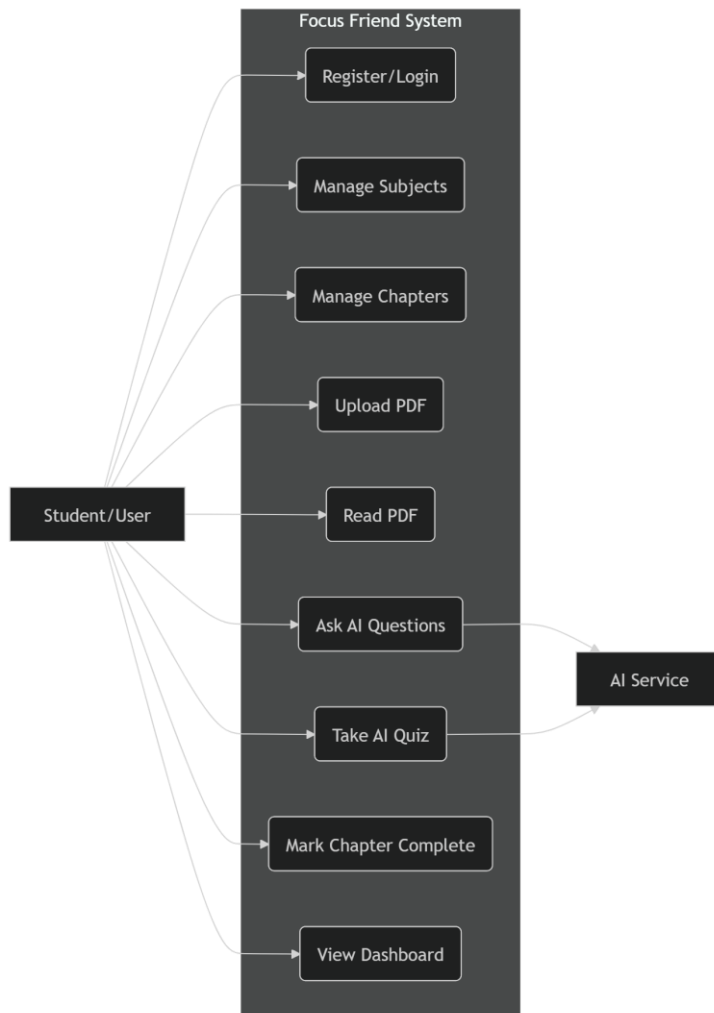
7. Use Case Diagram

The Use Case Diagram represents the interaction between the user (student) and the Focus Friend system. It shows how different functionalities of the system are accessed and used by the user and how the AI service supports the system. The diagram helps in understanding the system behavior from the user's perspective and identifies the main operations performed within the application.

The Focus Friend system mainly involves two actors:

- Student/User (Primary Actor)
- AI Service (Supporting Actor)

The use case diagram describes how the student interacts with the system to manage subjects, upload documents, ask questions, take quizzes, and track progress.



7.1 Actors in the System

7.1.1 Student/User (Primary Actor)

The Student is the main actor of the system. The student interacts directly with the application to perform all major activities such as registration, subject creation, document upload, asking questions, and taking quizzes.

Responsibilities of the Student:

- Register and login into the system
- Create and manage subjects
- Manage chapters
- Upload study PDFs
- Read documents
- Ask questions from AI
- Attempt quizzes
- Mark chapters complete
- Monitor dashboard progress

The entire system is designed primarily to serve the needs of the student.

7.1.2 AI Service (Secondary Actor)

The AI Service acts as a supporting actor. It provides intelligent responses based on the uploaded documents and helps generate quizzes.

Responsibilities of AI Service:

- Process user questions
- Generate contextual answers
- Generate quiz questions
- Assist in knowledge validation
- Support learning assistance features

The AI does not directly interact with the user interface but works through backend integration.

7.2 Main Use Cases of the System

The main use cases identified in the Focus Friend system are:

- User Registration
- User Login
- Subject Management
- Chapter Management
- PDF Upload
- PDF Reading
- AI Question Answering
- Quiz Attempt
- Chapter Completion
- Dashboard Monitoring

These use cases represent the core functionality of the system.

7.3 Use Case Description Table

Use Case	Actor	Description
Register User	Student	Allows a new user to create an account
Login User	Student	Allows existing user to access the system
Manage Subjects	Student	Create, update, and delete subjects
Manage Chapters	Student	View and manage chapter details
Upload PDF	Student	Upload chapter-specific study material
Read PDF	Student	View uploaded document inside system
Ask AI Questions	Student	Ask questions related to uploaded PDF
Generate Quiz	AI Service	Generate questions from document
Attempt Quiz	Student	Answer quiz questions
Evaluate Quiz	System	Check answers and calculate score
Mark Chapter Complete	System	Mark chapter after passing quiz
View Dashboard	Student	View progress statistics

7.4 Use Case Diagram Explanation

The use case diagram shows that the student must first register and log in to access the system. After authentication, the student can create subjects and chapters. The student can then upload PDFs related to chapters.

Once a PDF is uploaded, the student can perform two main actions:

- Ask questions from the AI assistant
- Attempt quizzes generated from the document

The AI service supports these operations by generating answers and quiz questions. After successfully passing the quiz, the system allows the student to mark the chapter as complete. Finally, the dashboard displays the updated progress.

The diagram shows a logical flow where:

Authentication → Study Management → Document Interaction → Knowledge Testing
→ Progress Tracking

This ensures that learning is structured and validated.

7.5 Use Case Diagram Description (Text Representation)

The Use Case Diagram can be summarized as:

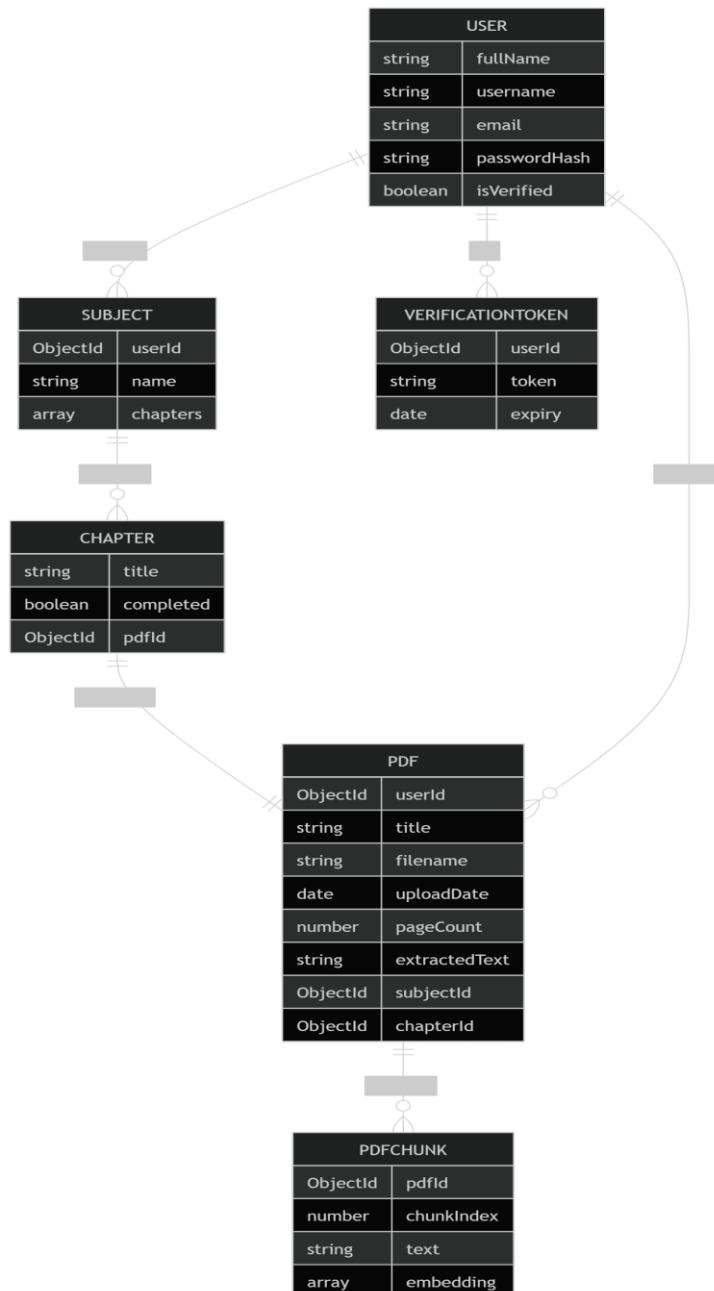
Student → Register/Login
Student → Manage Subjects
Student → Manage Chapters
Student → Upload PDF
Student → Read PDF
Student → Ask AI Questions
AI Service → Generate Answers
AI Service → Generate Quiz
Student → Attempt Quiz
System → Evaluate Quiz
System → Mark Chapter Complete
Student → View Dashboard

This shows the complete interaction between actors and system functions.

8. ER Diagram

The Entity Relationship (ER) Diagram represents the database structure of the Focus Friend system. It shows how different data entities such as users, subjects, chapters, PDFs, and tokens are related to each other. The ER diagram helps in designing the database schema and understanding how data flows inside the system.

The system uses MongoDB as the database, where each entity is represented as a collection and relationships are maintained using references. The ER design ensures proper data organization, integrity, and efficient retrieval.



8.1 Entities of the System

The main entities identified in the Focus Friend system are:

- User
- Subject
- Chapter
- PDF
- PdfChunk
- VerificationToken

Each entity stores specific information required for system functionality.

8.2 Entity Description

8.2.1 User Entity

The User entity stores information about registered students. It is the main entity because all other entities are linked to the user.

Attributes of User Entity:

Attribute	Description
user_id	Unique identifier of the user
name	Name of the student
email	User email address
password	Encrypted password
verified	Email verification status
createdAt	Account creation date

Purpose:

- Store user identity

- Manage authentication
- Link subjects with users

8.2.2 Subject Entity

The Subject entity stores subject information created by the user.

Attributes of Subject Entity:

Attribute	Description
subject_id	Unique subject ID
subject_name	Name of the subject
user_id	Reference to user
chapter_count	Total chapters
createdAt	Creation date

Purpose:

- Organize study topics
- Connect chapters to subjects
- Track subject-level progress

8.2.3 Chapter Entity

The Chapter entity represents individual study units inside subjects.

Attributes of Chapter Entity:

Attribute	Description
chapter_id	Unique chapter ID
chapter_name	Chapter title
subject_id	Reference to subject

Attribute	Description
pdf_id	Linked PDF
completed	Completion status
quiz_score	Quiz performance
createdAt	Creation date

Purpose:

- Represent study units
- Track completion
- Link documents with chapters

8.2.4 PDF Entity

The PDF entity stores information about uploaded study documents.

Attributes of PDF Entity:

Attribute	Description
pdf_id	Unique document ID
file_name	Name of PDF
file_path	Storage location
chapter_id	Linked chapter
upload_date	Upload timestamp
total_pages	Number of pages

Purpose:

- Store document metadata
- Connect documents to chapters
- Support document retrieval

8.2.5 PdfChunk Entity

The Pdf Chunk entity stores extracted text segments from the PDF. This is required for semantic search and AI interaction.

Attributes of Pdf Chunk Entity:

Attribute	Description
Chunk id	Unique chunk ID
Pdf id	Reference to PDF
Page number	Page number
Text content	Extracted text
Chunk index	Position of chunk

Purpose:

- Store extracted text
- Enable AI search
- Support context retrieval

8.2.6 Verification Token Entity

The Verification Token entity is used for email verification and account validation.

Attributes of Verification Token:

Attribute	Description
Token id	Unique token ID
User id	Reference to user
token	Verification token
expiry	Token expiry time

Purpose:

- Email verification
 - Account security
 - Token validation
-

8.3 Relationships Between Entities

The relationships between entities define how data is connected.

Main Relationships:

User → Subject (One to Many)

One user can create multiple subjects.

Subject → Chapter (One to Many)

One subject can contain multiple chapters.

Chapter → PDF (One to One)

Each chapter can have one associated PDF.

PDF → Pdf Chunk (One to Many)

One PDF contains multiple extracted text chunks.

User → PDF (One to Many)

One user can upload multiple PDFs.

8.4 Relationship Table

Entity 1 Relationship		Entity 2	Type
User	Creates	Subject	One-to-Many
Subject	Contains	Chapter	One-to-Many
Chapter	Linked with	PDF	One-to-One
PDF	Contains	Pdf Chunk	One-to-Many

Entity 1	Relationship	Entity 2	Type
User	Uploads	PDF	One-to-Many
User	Has	Verification Token	One-to-One

8.5 ER Diagram Explanation

The ER diagram shows that the User entity is the central entity. All subjects belong to a user, and each subject contains multiple chapters. Each chapter can have one PDF document attached. The PDF is further divided into multiple chunks to enable efficient AI search.

The verification token entity ensures account verification and security. This structure ensures that the system remains modular and scalable.

The ER diagram ensures:

- Data consistency
 - Efficient data retrieval
 - Proper relationship mapping
 - Scalable database design
 - Clear data hierarchy
-

8.6 Advantages of ER Design

The ER design provides several benefits:

- Clear database structure
- Avoids data redundancy
- Improves data consistency
- Supports scalability
- Simplifies backend development
- Improves query performance

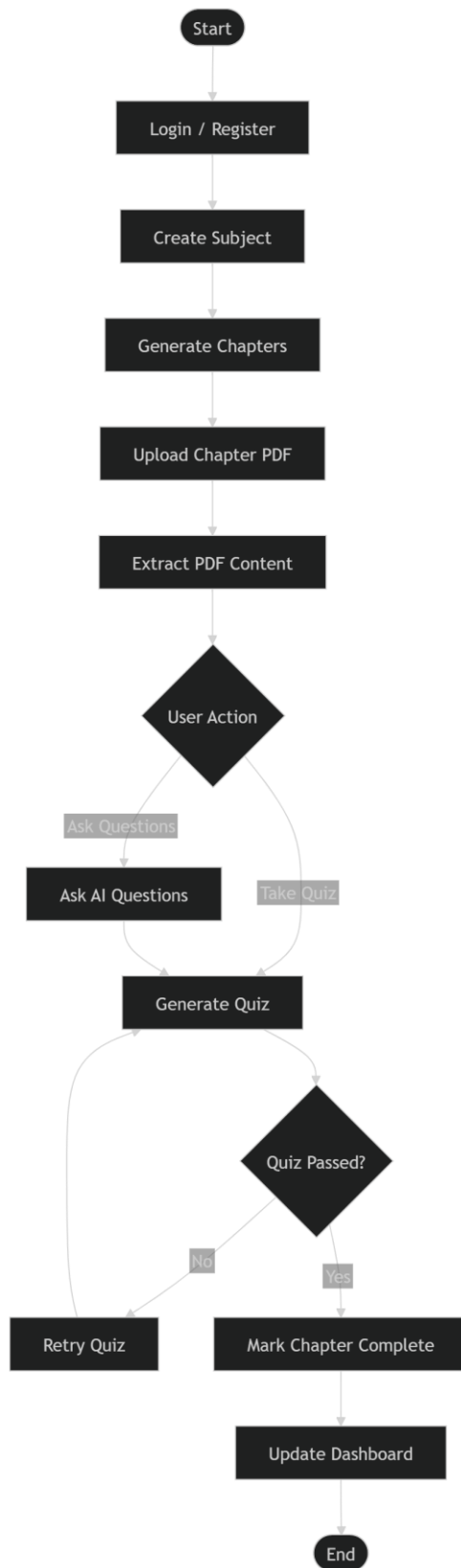
This structured database design helps the Focus Friend system manage academic data efficiently.

9. Flow Diagram

The Flow Diagram represents the overall workflow of the Focus Friend system. It shows the sequence of operations performed by the user and the system from login to chapter completion and progress tracking. The diagram helps in understanding how data flows through the system and how different modules interact with each other to complete the learning process.

The system follows a structured workflow to ensure that students follow a proper learning path instead of randomly accessing study materials. The workflow ensures that a chapter is only marked as complete after proper knowledge validation.

9.1 System Workflow Overview



The overall system flow can be summarized as:

User Registration/Login

- **Subject Creation**
- **Chapter Generation**
- **PDF Upload**
- **Text Extraction**
- **AI Question Answering / Quiz Generation**
- **Quiz Evaluation**
- **Chapter Completion**
- **Dashboard Update**

This workflow ensures structured learning and progress monitoring.

9.2 Stepwise Workflow Explanation

Step 1: User Registration and Login

The process starts with the user registering into the system by providing required details such as email and password. After successful registration, the user logs in using valid credentials.

The system verifies the credentials and generates a JWT token for secure access. Only authenticated users can access system features.

Process involved:

- User inputs credentials
 - System validates data
 - Password verification
 - JWT token generation
 - User session creation
-

Step 2: Subject Creation

After login, the user creates subjects based on their academic courses. The subject acts as a container for chapters.

Process involved:

- User enters subject name

- User enters number of chapters
- System creates subject record
- System auto-generates chapters

This step organizes study structure.

Step 3: Chapter Generation

Once a subject is created, chapters are automatically generated. The user can rename chapters if needed.

Process involved:

- Chapter records created
- Chapter linked with subject
- Chapter status initialized as incomplete

This prepares the study units.

Step 4: PDF Upload

The user uploads study material related to a chapter. The system accepts PDF files and stores them on the server.

Process involved:

- User selects PDF
- File validation performed
- File stored on server
- Metadata stored in database
- PDF linked with chapter

This step connects study material with chapters.

Step 5: Text Extraction and Processing

After uploading the PDF, the system extracts text using a PDF parsing library. The extracted content is divided into manageable chunks for AI processing.

Process involved:

- PDF reading
- Text extraction
- Page separation
- Chunk creation
- Database storage

This prepares data for AI interaction.

Step 6: AI Question Answering

After processing, the user can ask questions related to the uploaded document. The system searches relevant chunks using semantic similarity and sends them to the AI model.

Process involved:

- User asks question
- Relevant content retrieved
- Context prepared
- AI generates answer
- Response shown to user

This provides contextual learning support.

Step 7: Quiz Generation

To validate knowledge, the system generates quiz questions from the uploaded PDF.

Process involved:

- AI generates questions
- Questions displayed to user
- User submits answers
- System evaluates answers

This ensures active learning.

Step 8: Quiz Evaluation

The system checks whether the student answered correctly. If the student scores full marks, the chapter can be marked as complete. Otherwise, the student must retry.

Process involved:

- Answer comparison
- Score calculation
- Result generation
- Retry option if failed

This ensures knowledge validation.

Step 9: Chapter Completion

After successful quiz completion, the chapter status changes from incomplete to completed.

Process involved:

- Chapter status update
- Completion stored in database
- Progress recalculated

This ensures progress accuracy.

Step 10: Dashboard Update

Finally, the dashboard updates progress statistics.

Displayed information:

- Total subjects
- Total chapters
- Completed chapters
- Completion percentage

This provides visual feedback to the student.

9.3 Flow Control Logic

The system follows conditional flow logic:

If user authenticated → Allow access
If PDF uploaded → Enable AI interaction
If quiz passed → Allow completion
If quiz failed → Retry required
If chapter completed → Update dashboard

This ensures proper workflow control.

9.4 Benefits of Flow Design

The system flow design provides several advantages:

- Structured learning process
- Clear workflow sequence
- Knowledge validation
- Progress monitoring
- Reduced study confusion
- Better academic discipline

This workflow ensures that the system not only manages study material but also ensures effective learning.

9.5 Flow Diagram Description

The system flow can be described as:

Start
→ Login/Register
→ Create Subject
→ Generate Chapters
→ Upload PDF
→ Extract Text
→ Ask AI Questions
→ Generate Quiz

→ Attempt Quiz

→ Pass Quiz?

If YES → Mark Chapter Complete

If NO → Retry Quiz

→ Update Dashboard

→ End

This step-by-step workflow ensures a disciplined learning process.

10. Hardware and Software Requirements

The successful implementation of the Focus Friend system requires certain hardware and software components to ensure smooth operation, development, and deployment. These requirements ensure that the system can handle document processing, AI interaction, and database operations efficiently.

The requirements are divided into two categories:

- Hardware Requirements
 - Software Requirements
-

10.1 Hardware Requirements

The hardware requirements specify the minimum system configuration needed to develop and run the application efficiently.

Hardware Component	Minimum Requirement	Recommended Requirement
Processor	Intel i3 or equivalent	Intel i5 or higher
RAM	4 GB	8 GB or higher
Storage	500 MB free space	2 GB or more
Internet	Required	High speed broadband
Display	Standard Monitor	Full HD Display

Hardware Component	Minimum Requirement	Recommended Requirement
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Internet	Required	High speed broadband
Display	Standard Monitor	Full HD Display

Hardware Component Explanation

Processor:

A minimum Intel i3 processor is required to handle backend operations such as PDF processing and AI request handling. A higher processor improves performance.

RAM:

At least 4 GB RAM is required to run development tools such as Node.js, MongoDB, and code editors. More RAM improves multitasking ability.

Storage:

Storage is required for storing uploaded PDFs, application files, and database data. More storage is beneficial for handling larger document collections.

Internet Connection:

An internet connection is required for accessing AI APIs and downloading required dependencies.

Display:

A standard display is sufficient, but a higher resolution display improves development experience.

10.2 Software Requirements

The software requirements include the development tools, frameworks, programming languages, and runtime environments needed to build and execute the system.

Software	Purpose
Operating System	Platform to run the system
Node.js	Backend runtime
Express.js	Backend framework
MongoDB	Database management
Mongoose	Database modeling
HTML	Frontend structure
CSS	Frontend styling
JavaScript	Frontend logic
Google Gemini API	AI integration
Multer	File upload handling
PDF Parsing Library	Document processing

Software	Purpose
JWT Library	Authentication
VS Code	Development environment
Web Browser	Application testing

10.3 Software Component Explanation

Operating System

The system can run on Windows, Linux, or macOS. Cross-platform compatibility ensures flexibility during development and deployment.

Node.js

Node.js is used as the backend runtime environment. It enables server-side scripting and API development.

Express.js

Express.js is used to build REST APIs and manage routing, middleware, and request handling.

MongoDB

MongoDB is used as the primary database for storing user data, subjects, chapters, and extracted PDF content.

Mongoose

Mongoose is used as an Object Data Modeling (ODM) library to structure MongoDB data using schemas.

Frontend Technologies

HTML provides structure, CSS provides styling, and JavaScript provides interactivity in the user interface.

Google Gemini API

This API provides AI capabilities such as answering questions and generating quizzes based on uploaded documents.

Multer

Multer is used for handling multipart form data and managing file uploads.

PDF Parsing Library

Used for extracting text from uploaded PDF files to support AI interaction.

JWT (JSON Web Token)

Used for secure authentication and session management.

VS Code

Visual Studio Code is used as the primary development environment due to its extensions and debugging support.

Browser

Modern browsers such as Chrome, Edge, or Firefox are used for testing and accessing the application.

10.4 Development Environment Requirements

The development environment requires the following setup:

- Node.js installed
- MongoDB installed locally or cloud instance
- Code editor installed (VS Code recommended)
- Browser for testing
- Internet for AI integration

This setup allows developers to run and test the application locally.

10.5 System Compatibility

The system is designed to be compatible with:

Operating Systems:

- Windows
- Linux
- macOS

Browsers:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

This ensures accessibility across multiple platforms.

10.6 Advantages of Selected Technology Stack

The selected technology stack provides several advantages:

- Node.js provides fast backend performance
- MongoDB provides flexible data storage
- Express.js simplifies API development
- AI integration improves learning capability
- JavaScript enables full stack development
- JWT improves security
- Modular architecture supports scalability

These technologies were selected to ensure performance, flexibility, and ease of development.

10.7 Summary of Requirements

The Focus Friend system requires moderate hardware resources and commonly available software technologies. This makes the system easy to deploy and accessible to most students and developers. The use of open-source technologies also reduces development cost and improves flexibility.

11. Expected Outcome

The expected outcome of the Focus Friend project is the development of a fully functional academic productivity platform that helps students organize their study materials, improve their understanding of academic subjects, and track their learning progress effectively. The system is expected to transform traditional study methods into a more structured and measurable learning process by integrating document management, artificial intelligence assistance, and knowledge validation mechanisms.

The system is designed to provide a centralized environment where students can manage subjects, chapters, and study materials in one place. Instead of using multiple tools for document storage, doubt solving, and progress tracking, students will be able to perform all these activities within a single platform.

11.1 Academic Organization Improvement

The system is expected to improve the way students organize their study resources. By allowing users to create subjects and chapters, the platform provides a structured way to manage academic content.

Expected improvements include:

- Better organization of study materials
- Clear chapter structure
- Easy navigation of subjects
- Reduced confusion in study planning
- Centralized academic resource management

This will help students maintain discipline in their study routine.

11.2 Improved Learning Efficiency

The AI-based question answering system is expected to improve learning efficiency by providing instant answers from the uploaded study material. Instead of searching through entire documents, students can directly ask questions and receive relevant answers.

Expected benefits:

- Faster doubt resolution

- Context-based answers
- Reduced time spent searching notes
- Better concept clarity
- Personalized academic assistance

This feature promotes smart learning techniques.

11.3 Knowledge Validation Through Quiz System

One of the most important expected outcomes is knowledge validation. The system ensures that students do not just read study material but also demonstrate understanding through quizzes.

Expected outcomes include:

- Improved knowledge retention
- Active learning habits
- Self-assessment capability
- Mastery-based chapter completion
- Reduced passive learning

This helps students verify their understanding before moving to the next topic.

11.4 Progress Monitoring and Analytics

The dashboard system is expected to provide measurable academic progress indicators. Students will be able to see how much of their syllabus they have completed.

Expected benefits:

- Clear progress visualization
- Motivation through completion tracking
- Identification of incomplete chapters
- Improved time management
- Performance awareness

This helps students stay motivated and accountable.

11.5 Secure Academic Platform

The system is expected to provide a secure environment through authentication and data protection measures.

Security outcomes include:

- Secure login system
- Protected user data
- Safe document storage
- Controlled access to resources
- Protection against unauthorized use

This ensures trust and reliability.

11.6 Technical Learning Outcome

From a technical perspective, the project demonstrates the integration of multiple modern technologies into a single application.

Technical outcomes include:

- Practical implementation of full stack development
- Integration of artificial intelligence APIs
- Implementation of database design
- Secure authentication implementation
- Document processing techniques
- REST API development

This makes the project valuable from a software engineering learning perspective.

11.7 User Experience Improvement

The system is expected to improve the overall user experience by providing a simple and interactive interface.

Expected improvements:

- Easy navigation

- Interactive document reading
- Simple dashboard design
- Clear workflow
- Better usability

This ensures that students can use the system without technical difficulty.

11.8 Long-Term Educational Impact

In the long term, the system is expected to promote better study habits and structured learning practices. Students using the system may develop better academic discipline and learning strategies.

Long-term outcomes include:

- Improved study planning
- Better exam preparation
- Increased academic productivity
- Development of self-learning habits
- Technology-assisted education

This shows the educational value of the system.

11.9 Summary of Expected Outcome

Overall, the Focus Friend system is expected to deliver the following results:

- Organized academic study management
- AI-supported learning assistance
- Knowledge validation through quizzes
- Structured progress tracking
- Secure user management
- Improved learning discipline
- Practical implementation of modern technologies

The project demonstrates how artificial intelligence and web development technologies can be combined to create an effective academic productivity tool that enhances both learning efficiency and academic performance.

12. Limitations

Although the Focus Friend system provides many useful features for academic productivity and AI-assisted learning, there are certain limitations due to technical constraints, development scope, and time restrictions. These limitations identify areas where the system can be improved in the future.

Understanding these limitations is important because it helps in planning future improvements and scaling the system for real-world deployment.

12.1 Dependency on External AI Services

The system currently depends on external AI services (Google Gemini API) for question answering and quiz generation. If the AI service becomes unavailable or experiences downtime, some features of the application may not function properly.

Limitations include:

- AI features require internet connectivity
- API request limits may affect usage
- Service downtime can affect functionality
- API cost may increase in large-scale deployment

This dependency makes the system partially dependent on third-party services.

12.2 Limited Scalability of Document Search

Currently, the system performs semantic search using application memory and basic text matching techniques. While this works efficiently for small-scale use, performance may decrease if a large number of documents are uploaded.

Limitations include:

- Slower search for large datasets
- No vector database implementation
- Limited optimization for large-scale retrieval
- No distributed document indexing

Future improvements may include vector databases such as Pinecone or FAISS.

12.3 Local File Storage Limitation

Currently, uploaded PDFs are stored locally on the server. This approach is suitable for small projects but may not be ideal for production environments.

Limitations include:

- Storage limitations
- Risk of data loss without backup
- No cloud storage integration
- Limited scalability

Cloud storage solutions such as AWS S3 or Google Cloud Storage could solve this limitation.

12.4 Limited Account Management Features

The current system focuses mainly on study management and does not yet include complete account management features.

Missing features include:

- Forgot password functionality
- Password reset system
- Profile editing
- Account recovery options
- Two-factor authentication

These features could improve system security and usability.

12.5 Limited Testing and Deployment Features

Due to project scope limitations, the system does not include full production-level testing and deployment mechanisms.

Current limitations:

- No automated testing framework
- Limited performance testing
- No CI/CD pipeline
- No production monitoring tools
- No containerization (Docker)

Future versions could include DevOps practices.

12.6 Basic User Interface Design

The current system focuses more on functionality rather than advanced UI/UX design.

Limitations include:

- Basic interface design
- Limited accessibility features
- No theme customization
- Limited mobile responsiveness

Future improvements could include modern UI frameworks and responsive design.

12.7 Limited Quiz Complexity

The current quiz system generates basic questions from the document and evaluates answers in a simple manner.

Limitations include:

- Limited question types
- No adaptive difficulty
- No detailed performance analytics
- No timed assessments

Future improvements could include advanced quiz analytics.

12.8 Security Limitations

Although basic security measures are implemented, the system does not yet include advanced security features expected in production applications.

Current limitations:

- No two-factor authentication
- No OAuth login
- Limited intrusion detection
- No advanced encryption strategies
- No security audit logging

Future improvements could include advanced cybersecurity implementations (which would align well with your **Cyber Security specialization** ).

12.9 Integration Limitations

Currently, the system works as a standalone application and does not integrate with external academic platforms.

Missing integrations:

- No LMS integration
- No Google Classroom integration
- No calendar integration
- No notification system

Future integration could improve usability.

12.10 Summary of Limitations

The main limitations of the Focus Friend system are:

- Dependency on external AI services
- Limited scalability for large datasets
- Local file storage approach
- Limited account management features
- Lack of production-level testing

- Basic UI design
- Limited quiz functionality
- Basic security implementation

Despite these limitations, the system successfully demonstrates the core concept of AI-assisted academic productivity and provides a strong foundation for future development.

13. Conclusion

The Focus Friend project successfully demonstrates the development of an intelligent academic productivity system that integrates study management, document interaction, artificial intelligence assistance, and knowledge validation into a single platform. The system addresses common problems faced by students such as poor organization of study materials, lack of progress tracking, passive learning habits, and the absence of personalized academic assistance.

The project shows how modern web technologies such as Node.js, Express.js, MongoDB, and JavaScript can be combined with artificial intelligence services to build a practical real-world application. The implementation of PDF document processing and AI-based question answering demonstrates how students can interact more effectively with their study materials.

One of the key strengths of the system is the quiz-based chapter completion mechanism. Unlike traditional study management systems, Focus Friend ensures that chapters are only marked complete after the student demonstrates understanding through quizzes. This promotes active learning and mastery-based progress instead of passive reading.

From a technical perspective, the project demonstrates important software engineering concepts such as modular architecture, REST API development, database design, document processing, and secure authentication. The use of JWT authentication, password hashing, and validation techniques also shows the importance of implementing basic cybersecurity practices in modern applications.

The project also provides significant educational value by showing how artificial intelligence can be used as a learning assistant rather than just a general-purpose tool. By providing answers directly from uploaded study material, the system improves learning efficiency and helps students focus on relevant academic content.

Overall, the Focus Friend system achieves its objective of creating a structured and intelligent study management platform. The project successfully demonstrates how technology can be used to improve academic productivity and learning effectiveness. It also provides a strong foundation for future improvements and real-world deployment.

14. Future Work

Although the Focus Friend system provides essential features for academic productivity, several improvements can be implemented in future versions to enhance functionality, scalability, security, and user experience.

14.1 Advanced Account Management

Future versions can include additional account management features such as:

- Forgot password functionality
- Password reset through email verification
- Profile editing options
- Two-factor authentication (2FA)
- OAuth login (Google/GitHub login)

These improvements would enhance usability and security.

14.2 Improved AI Capabilities

The AI assistant can be further improved by adding:

- Source citations from PDF pages
- AI-generated summaries
- Flashcard generation
- Concept explanations
- Topic-wise study recommendations

These features would further improve learning effectiveness.

14.3 Vector Database Integration

To improve document retrieval performance, vector databases could be implemented.

Possible improvements:

- Vector embeddings for documents
- Faster semantic search
- Scalable document indexing
- Improved AI context retrieval

Technologies such as FAISS, Pinecone, or Weaviate could be used.

14.4 Cloud Storage Integration

Future systems can replace local PDF storage with cloud-based storage.

Possible improvements:

- AWS S3 integration
- Google Cloud Storage
- Secure document backup
- Scalable storage management

This would improve reliability and scalability.

14.5 Advanced Dashboard Analytics

The dashboard can be enhanced with more detailed analytics such as:

- Subject-wise performance graphs
- Quiz performance trends
- Study time tracking
- Learning recommendations
- Weak topic identification

This would provide better academic insights.

14.6 User Interface Improvements

Future UI improvements may include:

- Modern UI frameworks
- Dark mode support
- Mobile responsive design
- Accessibility improvements
- Customizable themes

This would improve user experience.

14.7 Testing and Deployment Enhancements

Future improvements could include:

- Automated testing
- Unit testing
- Integration testing
- CI/CD pipelines
- Docker containerization
- Cloud deployment

This would make the system production-ready.

14.8 Advanced Security Enhancements

Future security improvements may include:

- Two-factor authentication
- Role-based access control
- Activity logging
- API security monitoring
- Encryption of sensitive data
- Security audits

This is especially important for building a production-grade secure system (and aligns strongly with cybersecurity best practices).

14.9 System Integration Features

Future versions could integrate with:

- Learning Management Systems (LMS)
- Google Classroom
- Calendar reminders
- Notification systems
- Email alerts

This would make the system more practical for real-world academic use.

14.10 Final Future Scope Summary

The future scope of Focus Friend includes improvements in:

- AI intelligence
- Security features
- Scalability
- User experience
- Cloud deployment
- Academic analytics

With these improvements, Focus Friend could evolve from a minor academic project into a full-scale educational productivity platform.

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